HAOCHEN TIAN

Ph.D. Student ⋄ NLPR · MAIS · CASIA
Email: tianhaochen2023@ia.ac.cn ⋄ GitHub: github.com/hctian713 ⋄ Google Scholar

RESEARCH INTERESTS

Physical AI End-to-End Policy, World Model, Autonomous Driving, Robotics
MultiModal AI Unified Model, Vision-Language Model, Alignment, Evaluation

EDUCATION

Institute of Automation, Chinese Academy of Sciences Sep. 2023 - Present
Ph.D. in Computer Science & Technology, Advisor Prof. Liang Wang, Tieniu Tan

Wuhan University Sep. 2019 - Jun. 2023
B.S. in Spatial Informatics & Digital Technology

SELECTED PUBLICATIONS

∗ Equal Contribution † Project Lead ✉ Corresponding Author

Physical AI

SimScale: Learning to Drive via Real-World Simulation at Scale
Haochen Tian, Tianyu Li, Haochen Liu, Jiazhi Yang, Guang Li, Junli Wang, Yinfeng Gao, Zhang Zhang, Liang Wang, Hangjun Ye, Tieniu Tan, Long Chen, Hongyang Li
CVPR 2026 Oral (Top 0.88%, 141/16092). Paper | Web | Code | Dataset | Oral Pre

WorldEngine: Towards the Era of Post-Training for Autonomous Driving
OpenDriveLab, Huawei, and **Haochen Tian** as Contributor
Tech Report 2026. Paper | Web | Code

OneVL: One-Step Latent Reasoning and Planning with Vision-Language Explanation
Xiaomi Embodied Intelligence Team, and **Haochen Tian** as Contributor
Tech Report 2026. Paper | Web | Code

PlannerRFT: Reinforcing Diffusion Planners through Closed-Loop and Sample-Efficient Fine-Tuning
Hongchen Li, Tianyu Li, Jiazhi Yang, **Haochen Tian**, Caojun Wang, Lei Shi, Mingyang Shang, Zengrong Lin, Gaoqiang Wu, Zhihui Hao, Xianpeng Lang, Jia Hu, Hongyang Li
CVPR 2026. Paper | Web

Reinforced Refinement with Self-Aware Expansion for End-to-End Autonomous Driving
Haochen Liu, Tianyu Li, Haohan Yang, Li Chen, Caojun Wang, Ke Guo, **Haochen Tian**, Hongchen Li, Hongyang Li, Chen Lv
TPAMI 2026. Paper

DriveFine: Refining-Augmented Masked Diffusion VLA for Precise and Robust Driving
Chenxu Dang, Sining Ang, Yongkang Li, **Haochen Tian**, Jie Wang, Guang Li, Hangjun Ye, Jie Ma, Long Chen, Yan Wang
ECCV 2026. Paper | Code

TakeVLA: Learning from Mistakes: Post-Training for Driving VLA with Takeover Data
Yinfeng Gao, Deqing Liu, Qichao Zhang, Yupeng Zheng, **Haochen Tian**, Guang Li, Hangjun Ye, Long Chen, Da-Wei Ding, Dongbin Zhao
Preprint 2026. Paper

MultiModal AI

A Survey of Unified Multimodal Understanding and Generation: Advances and Challenges
Yan Yang*, **Haochen Tian***†, Yang Shi*, Wulin Xie*, Yi-Fan Zhang✉, Yuhao Dong, Yibo Hu, Liang Wang, Ran He, Caifeng Shan, Chaoyou Fu✉, Tieniu Tan
Preprint 2025. Paper | Web

RealUnify: Do Unified Models Truly Benefit from Unification?
Haochen Tian as Core Contributor.
CVPR 2026. Paper | Code | Dataset

MME-RealWorld: Could Your Multimodal LLM Challenge High-Resolution Real-World Scenarios that are Difficult for Humans ?
Yi-Fan Zhang, Huanyu Zhang, **Haochen Tian**, Chaoyou Fu, Shuangqing Zhang, Junfei Wu, Feng Li, Kun Wang, Qingsong Wen, Zhang Zhang, Liang Wang, Rong Jin, Tieniu Tan
ICLR 2025. Paper | Web | Code | Dataset

MM-RLHF: The Next Step Forward in Multimodal LLM Alignment
Yi-Fan Zhang, Tao Yu, **Haochen Tian**, Chaoyou Fu, Peiyan Li, Jianshu Zeng, Wulin Xie, Yang Shi, Huanyu Zhang, Junkang Wu, Xue Wang, Yibo Hu, Bin Wen, Fan Yang, Zhang Zhang, Tingting Gao, Di Zhang, Liang Wang, Rong Jin, Tieniu Tan
ICML 2025. Paper | Web | Code | Dataset

BaseReward: A Strong Baseline for Multimodal Reward Model
Yi-Fan Zhang, Haihua Yang, Huanyu Zhang, Yang Shi, Zezhou Chen, **Haochen Tian**, Chaoyou Fu, Haotian Wang, Kai Wu, Bo Cui, Xu Wang, Jianfei Pan, Haotian Wang, Zhang Zhang, Liang Wang
ICLR 2026. Paper

How Well Do Models Follow Visual Instructions? VIBE: A Systematic Benchmark for Visual Instruction-Driven Image Editing
Huanyu Zhang, Xuehai Bai, Chengzu Li, Chen Liang, **Haochen Tian**, Haodong Li, Ruichuan An, Yifan Zhang, Anna Korhonen, Zhang Zhang, Liang Wang, Tieniu Tan
Preprint 2026. Paper | Web | Code

OPEN-SOURCED PROJECTS

AlpaSim: A Modular, Lightweight, and Data-Driven Research Simulator for Autonomous Driving
NVIDIA, and **Haochen Tian** as Contributor
Code | Challenge

EXPERIENCE

Xiaomi EV, Embodied Intelligence Team — Sep. 2025 - Present
*Research Intern, **Top Talent Program*** — *Beijing, China*

- Worked with Prof. Long Chen and Naiyan Wang
- Worked on scalable closed-loop training for end-to-end autonomous driving.

Shanghai AI Lab → The University of Hong Kong, OpenDriveLab — Jun. 2024 - Present
Research Intern — *Hong Kong SAR, China*

- Worked with Prof. Hongyang Li
- Worked on human feedback reward for end-to-end autonomous driving.

Megvii Research, Foundation Model Group — Feb. 2024 - Mar. 2024
Research Intern — *Beijing, China*

- Worked on 3D-Aware Vision-Language-Action model for end-to-end autonomous driving.

HONORS / AWARDS

- **Xiaomi Top Talent Program** — 2026
- Merit Student of University of Chinese Academy of Sciences — 2024
- Outstanding Graduate of Wuhan University — 2023
- **Wangzhizhuo Innovation Scholarship (Top 1%)** — 2023

- **Gold Award of "Internet+" Innovation Competition** 2022
- Merit Student of Wuhan University 2021/2022

ACADEMIC SERVICE

Competition Organizer
AlpaSim End-to-End Closed-Loop Challenge 2026

Workshop Organizer
CVPR 2026 Workshop on From Labs to Life: Embodied Intelligence in the Wild

Standards Working Group Member
IEEE Standard P3474: Human Intentions and AI Alignment in Autonomous Driving Agents

Conference/Journal Reviewer
CVPR, ECCV, NeurIPS, ICRA, RSS, WACV, BMCV, RA-L, T-PAMI, etc.

Invited Talk
Scalable Real-World Simulation Reshapes End-to-End Policy Learning THU | AD Heart | CV3D etc, Dec. 2025

SKILLS

Language
Chinese (Native) | English (Proficient CET6 500+, CET4 600+)

Programming
Python, C++, Matlab, Java, JavaScript, SQL